

Day 3 — Both Numbers Below 100

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit multiplication where both factors are between 90 and 99 using the Nikhilam method, in under 10 seconds.

Materials

- Whiteboard with the vertical-line template.
- Class wall-poster: “Deficits from 100” — small reference card (e.g. $97 \rightarrow 3$, $88 \rightarrow 12$).
- Pre-printed drill sheet (10 problems).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-100 lightning round.** Teacher calls 2-digit numbers in the 80s and 90s; class chants the deficit. ($97 \rightarrow 3$. $88 \rightarrow 12$. $96 \rightarrow 4$. $92 \rightarrow 8$. $99 \rightarrow 1$. $85 \rightarrow 15$)
- Repeat. By the end most students should be doing this in under a second.

Core (15 min) — The 2-digit $\times$ 2-digit case, both below 100

1. **The template** (vertical line, base = 100):

$$\begin{array}{r|l} 97 & -3 \\ \times 96 & -4 \\ \hline ? & ? \end{array} \quad \leftarrow \text{deficit from 100}$$

2. **Walk through 97×96** (from Day 1, now formalised):
 - LEFT: $97 - 4 = \mathbf{93}$.
 - RIGHT: $3 \times 4 = \mathbf{12}$.
 - **Pad to 2 digits** (base 100 has 2 zeros): RIGHT = 12 (already 2 digits, no pad needed).
 - Answer: **9312**.
3. **Walk through 98×99 :**
 - Deficits: 2, 1.
 - LEFT: $98 - 1 = \mathbf{97}$.
 - RIGHT: $2 \times 1 = 2 \rightarrow$ pad to 2 digits $\rightarrow \mathbf{02}$.
 - Answer: **9702**.
4. **The padding rule (write on board, boxed):**

Pad the right part to as many digits as there are zeros in the base. Base 10 → 1 digit on the right. Base 100 → 2 digits on the right. Base 1000 → 3 digits on the right.

5. **The overflow / carry case: 88×89 .**

- Deficits: 12, 11.
- LEFT: $88 - 11 = 77$. (Or $89 - 12 = 77$. ✓)
- RIGHT: $12 \times 11 = 132$.
- But the right part should be only 2 digits (base 100). 132 has 3 digits — the leading 1 overflows.
- **Carry the overflow into the left:** left = $77 + 1 = 78$; right = 32.
- Answer: **7832**. ✓ (Calculator check: $88 \times 89 = 7832$.)

6. **The “use either cross” check.** For 97×96 :

- Either $97 - 4 = 93$ or $96 - 3 = 93$.
- If you get *different* numbers, you’ve made an error. This is your built-in error-detector.

Activity (20 min) — Drill

Pair drill, then individual silent drill.

Drill sheet (10 problems, mixed difficulty):

- | | | | |
|-------------------|--------------------|-------------------|-------------------|
| 1. 97×96 | 2. 98×99 | 3. 95×94 | 4. 93×91 |
| 5. 99×98 | 6. 92×95 | 7. 88×89 | 8. 87×92 |
| 9. 96×96 | 10. 91×87 | | |

- **5 min individual:** solve all 10 silently. No talking.
- **5 min paired check:** swap sheets. Partner reads problem 1; the other resolves it on a fresh strip; compare. Continue.
- **10 min mixed work:** problems with both-below-100, individual silent.

Answer key (teacher): | Q | Deficits | Left | Right | Answer | | 1 | 3, 4 | 93 | 12 | 9312 | | 2 | 2, 1 | 97 | 02 | 9702 | | 3 | 5, 6 | 89 | 30 | 8930 | | 4 | 7, 9 | 84 | 63 | 8463 | | 5 | 1, 2 | 97 | 02 | 9702 | | 6 | 8, 5 | 87 | 40 | 8740 | | 7 | 12, 11 | $77+1=78$ | 32 | 7832 | | 8 | 13, 8 | $87-13=79$ or $92-8=84$... | (see note) | 8004 | | 9 | 4, 4 | 92 | 16 | 9216 | | 10 | 9, 13 | $91-13=78$ or $87-9=78$ | 117 → carry 1 → 79, right 17 | 7917 |

(Note Q8: $87 - 8 = 79$, OR $92 - 13 = 79$. Right = $13 \times 8 = 104$. Carry 1 → 80; right = 04. Answer: 8004.)

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading. Hand back tomorrow.

Student-facing content

Both numbers below 100

Step	What you do
1	Find each number's deficit from 100 .
2	Cross-subtract for the LEFT part.
3	Multiply the two deficits for the RIGHT part.
4	Pad the RIGHT part to 2 digits .
5	If the RIGHT part overflows (≥ 100), carry the overflow into the LEFT.

Examples: $-97 \times 96 \rightarrow 93 \mid 12 \rightarrow \mathbf{9312}$ $-98 \times 99 \rightarrow 97 \mid 02 \rightarrow \mathbf{9702}$ $-88 \times 89 \rightarrow 77+1 \mid 32 \rightarrow \mathbf{7832}$ (right part overflowed)

Sanskrit name: *Nikhilam Navataścaramaṁ Daśataḥ* — “all from 9 and the last from 10.”

Homework

- 10 problems on the back of today's worksheet, all 2-digit $\times$ 2-digit with both factors in 85–99.
- Time yourself. Beat the class average from today.

Differentiation

- Tier up:** Try 83×89 and 78×85 (further from 100). The method still works, but the deficits are larger and the right-part overflow is more common. See how far below 100 you can go before the method stops feeling fast.
- Tier down:** Stick to factors in 90–99 for the rest of the week. We'll revisit 80s on Day 5.

Teacher notes

- The overflow / carry is the single most common error point. Spend time on 88×89 and 78×85 . If students don't fluent on the carry by end of class, schedule 5 min of catch-up at the start of Day 4.
- The “use either cross” check is a real check — encourage students to do both crosses initially. By Day 6 they'll naturally drop the second one.
- Some students will be fast on a calculator and slow on Nikhilam. Acknowledge: “*With a calculator, calculator wins. The Nikhilam method matters when you DON'T have a calculator — exams, mental estimation, daily life.*”

Citation(s) used in this lesson

- Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, base-100 examples.